Reference	Qty	Description	Manufacturer
U1	1	8-bit AVR microcontroller, ATtiny412, SOIC-8	Microchip
C1	1	0.1 µF ±10% 25V Ceramic Capacitor X7R 0805 (2012 Metric)	Walsin
R1	1	10 kΩ, 1%, 0805 resistor	YAGEO
R2	1	100 kΩ, 1%, 0805 resistor	YAGEO
R3	1	1 MΩ, 1%, 0805 resistor	YAGEO
R4	1	1 kΩ, 1%, 0805 resistor	YAGEO
R5,R6	2	5.1 kΩ, 1%, 0805 resistor	Bourns
J1	1	1×2, 2.54 mm straight female socket	Same Sky
J2	1	1×4, 2.54 mm right-angle female socket	Sullins
J3	1	1×3, 2.54 mm straight male header	Amphenol FCI
J4	1	USB-C receptacle, power-only	GCT
S1	1	2-pole, 4-position rotary switch	Alpha Taiwan
LCD1	1	16×2 I²C LCD module	DFRobot
PCB-mounted parts subtotal			
Complete device subtotal (includes external LCD)			

Pricing note Enter current supplier unit prices in column I. Shipping, tax, PCB fabrication, and asse

Verification note This BOM is based on the selected parts recorded during design. Verify all MPNs, foot

Manufacturer Part Number
ATTINY412-SSNR
0805B104K250CT
RC0805FR-0710KL
RC0805FR-07100KL
AF0805FR-071ML
RC0805FR-071KL
CR0805-FX-5101ELF
SSK02-FPH-254-A2
PPTC041LGBN-RC
10129378-903001BLF
USB4085-GF-A
SR10010F-0204-20F0B-C7-N-0027
DFR0555

mbly are not included.

:prints, switch terminal mapping, connector

KiCad Footprint / Connection	Assembly
Package_SO:SOIC-8_3.9x4.9mm_P1.27mm	PCB-mounted
Capacitor_SMD:C_0805_2012Metric	PCB-mounted
Resistor_SMD:R_0805_2012Metric	PCB-mounted
Resistor_SMD:R_0805_2012Metric	PCB-mounted
Resistor_SMD:R_0805_2012Metric	PCB-mounted
Resistor_SMD:R_0805_2012Metric	PCB-mounted
Resistor_SMD:R_0805_2012Metric	PCB-mounted
Connector_PinSocket_2.54mm:PinSocket_1x02_P2.54mm_Vertical	PCB-mounted
Connector_PinSocket_2.54mm:PinSocket_1x04_P2.54mm_Horizontal	PCB-mounted
Connector_PinHeader_2.54mm:PinHeader_1x03_P2.54mm_Vertical	PCB-mounted
Connector_USB:USB_C_Receptacle_GCT_USB4085	PCB-mounted
switch:SR10010F020420F0BC7N0027	PCB-mounted
Not on PCB (external module)	External / off-board

r pin order, and current prices against manufacturer datasheets before ordering.

Notes	Unit Price (USD)
	$0.58
Verify voltage/tolerance against final schematic	$0.11
	$0.10
	$0.10
	$0.14
	$0.10
USB-C CC pulldown	$0.10
Test resistor input connector	$0.24
LCD connection	$0.50
UPDI/programming connector	$0.27
USB-C Power	$0.91
Estimated tarriff of $1.76	$5.86
Connects to J2; Is for reading the outputs	$9.90

Extended Price (USD)
$0.58
$0.11
$0.10
$0.10
$0.14
$0.10
$0.20
$0.24
$0.50
$0.27
$0.91
$5.86
$9.90
$9.11
$19.01

Website Supplier
https://www.digikey.com/en/products/detail/microchip-technology/ATTINY412-SSNR/8594943?s=N4IgTCBcDaI
https://www.digikey.com/en/products/detail/walsin-technology-corporation/0805B104K250CT/9355063
https://www.digikey.com/en/products/detail/yageo/RC0805FR-0710KL/727535?s=N4IgTCBcDaIEoGEAMAOJBW
https://www.digikey.com/en/products/detail/yageo/RC0805FR-07100KL/727544?s=N4IgTCBcDaIEoGEAMAOJB
https://www.digikey.com/en/products/detail/yageo/AF0805FR-071ML/5901272?s=N4IgTCBcDaIIDEAMAOJBW
https://www.digikey.com/en/products/detail/yageo/RC0805FR-071KL/727444?s=N4IgTCBcDaIEoGEAMAOJBWA
https://www.digikey.com/en/products/detail/bourns-inc/CR0805-FX-5101ELF/3784995?s=N4IgTCBcDaIMICUAM
https://www.digikey.com/en/products/detail/same-sky-formerly-cui-devices/SSK02-FPH-254-A2/28772215
https://www.digikey.com/en/products/detail/sullins-connector-solutions/PPTC041LGBN-RC/775898?s=N4IgTCBcDaIApwCoGEAMAWAjAGQOICEA5AWgCVkQBdAXyA
https://www.digikey.com/en/products/detail/amphenol-cs-fci/10129378-903001BLF/7915928?s=N4IgTCBcDaIl
https://www.digikey.com/en/products/detail/gct/USB4085-GF-A/9859662?s=N4IgTCBcDaIKoGUBCAWADADgKv
https://www.mouser.com/en/ProductDetail/Alpha-Taiwan/SR10010F-0204-20F0B-C7-N-0027?qs=8%252Br4Hz5
https://www.digikey.com/en/products/detail/dfrobot/DFR0555/9356340?s=N4IgTCBcDaICIDEBKAGArBkBdAvkA

IIBUEEkByBNALARjAWgGUDUAlEAXQF8g

WAYnAtEg7AIxJIDSAMiALoC%2BQA

/AOJBWAtAMQBqfQEYlCBRAGWxAF0BfIA

wAY5gJwGYDsAOAtChaCSAQgDIBiIAugL5A

5Xir%2Ffy%252Bqxa0GVpg%3D%3D